

PROMPT ENGINEERING

A practical student handbook with formulas, real-time examples, trainer notes and assignments

“A better prompt creates a better possibility.”

10 TECHNIQUES

30+ EXAMPLES

PRACTICE TASKS

Start Here: What Is Prompt Engineering?

Prompt engineering is the practice of writing clear, structured and effective instructions for an AI system so that it can produce a useful, relevant and correctly formatted response.

GOLDEN FORMULA

ROLE + CONTEXT + TASK + INSTRUCTIONS + CONSTRAINTS + OUTPUT FORMAT

The six building blocks

Role	Who should the AI act as?	Act as a Java trainer.
Context	What background does it need?	The learners are fresh graduates.
Task	What should it do?	Create a five-day training plan.
Instructions	What steps must it follow?	Add concepts, demos and assignments.
Constraints	What limits or rules apply?	Use simple English; below 500 words.
Format	How should it present the answer?	Return a six-column table.

Example of a complete professional prompt

PROMPT

Act as an experienced Java corporate trainer. I am teaching 30 fresh graduates with no programming experience. Create a five-day Java fundamentals plan. Include three topics, one demonstration, one hands-on activity and one assignment each day. Use Java 17 and simple English. Do not include Spring Boot or advanced concepts. Present the answer in a table and keep it below 500 words.

This combines role-based, context-based, instruction-based, constraint-based, negative and output-format prompting.

01 Zero-Shot Prompting

SIMPLE MEANING	Ask the AI to complete a task without showing any example. The model relies on its existing knowledge and the clarity of your instruction.
-----------------------	--

FORMULA	TASK + TOPIC + REQUIREMENT
----------------	----------------------------

Reusable prompt structure

TEMPLATE	[Action] + [topic/input] + [quality, audience or result needed]
-----------------	---

Real-time examples

Workplace email

PROMPT	Write a polite email to a customer apologising for the delayed delivery of a laptop. Mention that the revised delivery date is Monday.
---------------	--

Why it works: The task and required facts are explicit; no demonstration is needed.

Technical learning

PROMPT	Explain Java inheritance to a beginner using a simple family-property example.
---------------	--

Why it works: The audience and analogy guide the depth.

Travel

PROMPT	Create a two-day Hyderabad trip plan with three tourist places and one famous food each day.
---------------	--

Why it works: The result is common and can be generated directly.

When should you use it?

- For simple, direct and familiar tasks.
- When you need a quick first draft.
- When no special pattern or style must be copied.

ADVANTAGE	Fast, simple and requires very little preparation.
LIMITATION	The result can be generic when the task, audience or format is not specific.
COMMON MISTAKE	Weak: 'Tell me about Java.' Better: mention the learner, purpose, topics and output length.
TRAINER NOTE	Zero-shot does not mean zero detail. A good zero-shot prompt can still contain context, constraints and a format.

02 One-Shot Prompting

SIMPLE MEANING	Give the AI one complete example, then ask it to follow the same pattern for a new input.
-----------------------	---

FORMULA	ONE EXAMPLE + NEW INPUT + SAME PATTERN
----------------	--

Reusable prompt structure

TEMPLATE	Example: [input] → [desired output] Now: [new input] → ?
-----------------	--

Real-time examples

Sentiment classification

PROMPT	Example: 'The camera quality is excellent.' → Positive Now classify: 'The battery drains within two hours.' → ?
---------------	---

Why it works: The example demonstrates both the label and answer style.

Professional rewriting

PROMPT	Example: 'Send me the report.' → 'Could you please share the report at your earliest convenience?' Now rewrite professionally: 'Give me tomorrow's meeting details.'
---------------	--

Why it works: The AI copies the polite transformation.

Concept definition

PROMPT	Example: Java → Object-oriented programming language Now follow the same format: Python → ?
---------------	---

Why it works: A single sample establishes the expected short definition.

When should you use it?

- When one example clearly shows the desired pattern.
- For tone conversion, tagging or simple transformations.
- When a particular answer length or style is required.

ADVANTAGE	More predictable than zero-shot because the target pattern is visible.
------------------	--

LIMITATION	One example may be misleading or may not represent all possible cases.
-------------------	--

COMMON MISTAKE	Providing an example that does not match the actual task causes the AI to copy the wrong pattern.
-----------------------	---

TRAINER NOTE	Choose a clean, representative example. The quality of the demonstration strongly influences the answer.
---------------------	--

03 Few-Shot Prompting

SIMPLE MEANING	Give the AI two or more examples so it can learn a pattern, distinguish categories and respond consistently to a new input.
-----------------------	---

FORMULA	EXAMPLE 1 + EXAMPLE 2 + EXAMPLE 3 + NEW INPUT
----------------	---

Reusable prompt structure

TEMPLATE	[Input A] → [Output A] [Input B] → [Output B] [Input C] → [Output C] New input → ?
-----------------	--

Real-time examples

Feedback classification

PROMPT	'The application is easy to use.' → Positive 'The application crashes every day.' → Negative 'The application was updated yesterday.' → Neutral Now classify: 'The dashboard saves a lot of time.' → ?
---------------	--

Why it works: Examples define all three categories.

Student performance

PROMPT	90-100 → Excellent 75-89 → Very Good 50-74 → Good Below 50 → Needs Improvement Classify a learner who scored 82.
---------------	--

Why it works: The score bands remove ambiguity.

REST API understanding

PROMPT	GET /employees → Retrieve all employees POST /employees → Add a new employee DELETE /employees/10 → Delete employee 10 Explain: PUT /employees/10
---------------	---

Why it works: Several patterns help infer the missing operation.

When should you use it?

- Classification, extraction and consistent formatting.
- When categories may be confused.
- When zero-shot or one-shot answers are inconsistent.

ADVANTAGE	Usually improves accuracy, consistency and pattern following.
LIMITATION	Long or contradictory examples consume space and confuse the intended pattern.
COMMON MISTAKE	Using only positive examples when the task has positive, negative and neutral categories.
TRAINER NOTE	Cover the important variations. Keep demonstrations short, correct and formatted consistently.

04 Role-Based Prompting

SIMPLE MEANING	Assign the AI a profession, responsibility or perspective so its response uses an appropriate tone, depth and focus.
-----------------------	--

FORMULA	ROLE + TASK + AUDIENCE + EXPECTED RESULT
----------------	--

Reusable prompt structure

TEMPLATE	Act as a [role]. Perform [task] for [audience]. Include [requirements].
-----------------	---

Real-time examples

Mock interview

PROMPT	Act as an experienced Java interviewer. Conduct a mock interview for a developer with two years of experience. Ask one question at a time; after each answer, provide feedback and a score out of 10.
---------------	---

Why it works: The role shapes the questions and evaluation style.

Corporate training

PROMPT	Act as a senior corporate trainer. Create a five-day prompt-engineering workshop for software employees. Include concepts, demonstrations, exercises and daily assessments.
---------------	---

Why it works: The role encourages training-oriented structure.

Travel planning

PROMPT	Act as a local Kashmir travel guide. Build a five-day family itinerary starting from Srinagar, including travel time, food and approximate cost.
---------------	--

Why it works: The local-guide perspective emphasises practical travel details.

When should you use it?

- Expert-style explanation or feedback.
- Interviews, simulations, coaching and reviews.
- When tone and professional perspective matter.

ADVANTAGE	Improves domain focus, vocabulary, audience fit and tone.
LIMITATION	A role does not guarantee factual accuracy or professional certification.
COMMON MISTAKE	Writing only 'Act as an expert' without defining the expertise, task or audience.
TRAINER NOTE	Give the role a responsibility: 'Act as a reviewer who checks security, clarity and correctness.'

05 Context-Based Prompting

SIMPLE MEANING	Provide relevant background information so the AI understands the situation before completing the task.
-----------------------	---

FORMULA	BACKGROUND + CURRENT SITUATION + TASK + RESULT
----------------	--

Reusable prompt structure

TEMPLATE	Background: [facts] Situation: [need/problem] Task: [action] Expected result: [goal]
-----------------	--

Real-time examples

Training design

PROMPT	I am training 30 fresh graduates with no programming experience. We have five days and six hours per day. The objective is practical Java fundamentals. Create a plan covering variables, conditions, loops, arrays, methods and basic OOP.
---------------	---

Why it works: Learner level and available time determine the plan.

Resume summary

PROMPT	I have five years of Angular experience and two years in testing. I am applying for a Senior Frontend Developer role requiring TypeScript, RxJS and REST APIs. Write a suitable resume summary.
---------------	---

Why it works: The job context makes the summary targeted.

Customer support

PROMPT	A customer ordered a laptop five days ago. Delivery was promised in three days but was delayed by heavy rain. The revised date is Monday. Write a polite apology email.
---------------	---

Why it works: The background prevents invented or irrelevant details.

When should you use it?

- Personalised advice and targeted writing.
- Business or technical problem-solving.
- Whenever background facts materially affect the answer.

ADVANTAGE	Produces more relevant, realistic and personalised responses.
LIMITATION	Too much irrelevant background can hide the real task.
COMMON MISTAKE	Copying a large document without telling the AI which parts matter.
TRAINER NOTE	Use the 'need-to-know' test: include only facts that can change the response.

06 Instruction-Based Prompting

SIMPLE MEANING	Tell the AI exactly what actions or steps to perform. This is useful when completeness and sequence matter.
-----------------------	---

FORMULA	ACTION 1 + ACTION 2 + ACTION 3 + FINAL DELIVERABLE
----------------	--

Reusable prompt structure

TEMPLATE	Complete these steps: 1. [action] 2. [action] 3. [action] Return [final deliverable].
-----------------	---

Real-time examples

Code review

PROMPT	Review the Java code below. 1) Identify compilation errors. 2) Explain each error simply. 3) Correct the code. 4) Provide the final program. 5) Suggest two improvements. [Paste code]
---------------	--

Why it works: Numbered actions reduce missed requirements.

Article study

PROMPT	Read the article. 1) Identify the main topic. 2) Summarise it in five bullets. 3) List three key terms. 4) Create five questions. 5) Give a one-sentence conclusion.
---------------	--

Why it works: The workflow turns one article into several learning outputs.

REST lesson

PROMPT	Explain REST APIs. First define REST; next explain GET, POST, PUT and DELETE; then give one employee example for each; finally provide a practice task.
---------------	---

Why it works: The sequence moves from concept to application.

When should you use it?

- Multi-step analysis, reviews and transformations.
- When the order of work is important.
- When students need a structured learning process.

ADVANTAGE	Improves completeness, traceability and organisation.
LIMITATION	Conflicting or excessively detailed instructions may reduce quality.
COMMON MISTAKE	Mixing the task, data and rules in one long paragraph with no visible sequence.
TRAINER NOTE	Use numbered steps and end with a clear definition of the final deliverable.

07 Constraint-Based Prompting

SIMPLE MEANING	Set specific limits and rules for length, time, budget, technology, tone, audience or number of items.
-----------------------	--

FORMULA	TASK + MUST INCLUDE + LIMITS + RULES
----------------	--------------------------------------

Reusable prompt structure

TEMPLATE	Perform [task]. Must include [items]. Keep within [limit]. Follow [rules].
-----------------	--

Real-time examples

LinkedIn post

PROMPT	Write a LinkedIn post about completing Generative AI training. Use simple professional English, exactly three learning points, one tagline and five hashtags. Keep it below 100 words.
---------------	--

Why it works: Measurable constraints make the answer publication-ready.

Travel budget

PROMPT	Create a two-day Hyderabad plan for four people. Keep the total budget below Rs.10,000. Include exactly three places and one famous food each day. Use locations within 15 km of the city centre.
---------------	---

Why it works: Budget, distance and item counts control feasibility.

Java program

PROMPT	Create an employee-record program using only arrays, loops and methods. Do not use collections, databases or external libraries. Keep the code below 100 lines.
---------------	---

Why it works: Technology boundaries make it suitable for a particular lesson.

When should you use it?

- Strict business, assessment or publishing requirements.
- When time, cost, length or technology is limited.
- When the response must contain an exact number of items.

ADVANTAGE	Provides control and makes responses easier to evaluate.
LIMITATION	Too many or impossible constraints can harm usefulness and creativity.
COMMON MISTAKE	Using vague limits such as 'short', 'cheap' or 'nearby' instead of measurable values.
TRAINER NOTE	Prefer measurable wording: 'under 150 words', 'exactly five points', 'within Rs.5,000'.

08 Negative Prompting

SIMPLE MEANING	Tell the AI what it must avoid or exclude, while also explaining the positive result you want.
-----------------------	--

FORMULA	TASK + INCLUDE + DO NOT INCLUDE
----------------	---------------------------------

Reusable prompt structure

TEMPLATE	Perform [task]. Include [required items]. Do not include [unwanted items].
-----------------	--

Real-time examples

Beginner explanation

PROMPT	Explain artificial intelligence to a 10-year-old using one daily-life example. Do not use technical terminology, mathematics, code or complex definitions.
---------------	--

Why it works: Exclusions protect the learner from unsuitable complexity.

Birthday wish

PROMPT	Write a funny and respectful birthday wish for my friend. Do not mention age, marriage, salary or weight.
---------------	---

Why it works: Sensitive topics are clearly excluded.

Family travel

PROMPT	Create a family-friendly two-day Goa plan with beaches, food and sightseeing. Do not include pubs, casinos, nightlife or expensive resorts.
---------------	---

Why it works: Negative instructions keep recommendations appropriate.

When should you use it?

- Avoiding sensitive, unsafe or advanced content.
- Restricting technologies or themes.
- Removing styles, topics or assumptions that repeatedly appear.

ADVANTAGE	Keeps the response focused and prevents unwanted content.
------------------	---

LIMITATION	Only saying what not to do leaves the desired answer unclear.
-------------------	---

COMMON MISTAKE	Weak: 'Do not write a long answer.' Better: 'Use exactly three bullets, each under ten words.'
-----------------------	--

TRAINER NOTE	Pair every important exclusion with a positive direction: what to include, for whom and in what style.
---------------------	--

09 Output-Format Prompting

SIMPLE MEANING	Specify the exact structure of the final answer: table, bullets, email, JSON, code block, checklist or report.
-----------------------	--

FORMULA	TASK + REQUIRED INFORMATION + EXACT OUTPUT STRUCTURE
----------------	--

Reusable prompt structure

TEMPLATE	Perform [task]. Return only [format]. Use these [columns/fields/sections]: [...].
-----------------	---

Real-time examples

Comparison table

PROMPT	Compare Java and Python in a table with columns: Feature, Java, Python and Recommended Use. Compare syntax, performance, learning difficulty, applications and jobs.
---------------	--

Why it works: Named columns make the comparison scannable.

JSON data

PROMPT	Generate three Java course records. Return only valid JSON. Each object must contain courseName, duration, level and topics.
---------------	--

Why it works: The structure makes the output machine-readable.

Training plan

PROMPT	Create a five-day Java plan in a table with Day, Topics, Duration, Practical Program, Assignment and Learning Outcome.
---------------	--

Why it works: The format supports classroom use and evaluation.

When should you use it?

- Consistent reports, tables and reusable content.
- Data that will be consumed by software.
- When readers must compare many fields quickly.

ADVANTAGE	Makes output organised, readable, consistent and reusable.
LIMITATION	AI may occasionally add extra text or produce invalid syntax.
COMMON MISTAKE	Requesting 'table format' but not defining the columns or row meaning.
TRAINER NOTE	For strict integration, say 'Return only valid JSON; no Markdown or explanation' and validate the result.

10 Iterative Prompting

SIMPLE MEANING	Create a first version and then improve it through focused follow-up prompts. Each iteration changes one or more clear qualities.
-----------------------	---

FORMULA	INITIAL DRAFT → REVIEW → REFINE → VERIFY → FINALISE
----------------	---

Reusable prompt structure

TEMPLATE	Prompt 1: Create the draft. Prompt 2: Improve [specific quality]. Prompt 3: Add/remove [specific content]. Prompt 4: Apply final constraints and format.
-----------------	--

Real-time examples

Resume development

PROMPT	1) Write a resume summary for a Java full-stack developer with five years' experience. 2) Reduce it below 80 words. 3) Add Spring Boot, Microservices, Angular and AWS. 4) Target a Senior Software Engineer role. 5) Use a confident tone without exaggerated claims.
---------------	--

Why it works: Each turn has a focused improvement goal.

FastAPI development

PROMPT	1) Create an endpoint to add an employee. 2) Add Pydantic validation. 3) Connect MySQL. 4) Handle duplicate IDs. 5) Add comments and a beginner explanation.
---------------	--

Why it works: The application grows in testable stages.

Training content

PROMPT	1) Draft a five-day plan. 2) Add hands-on activities. 3) Adjust for beginners. 4) Limit each lecture to 45 minutes. 5) Convert the final plan to a table.
---------------	---

Why it works: Refinement aligns content, audience, timing and format.

When should you use it?

- Complex content, coding, planning and design.
- When the first answer needs targeted improvement.
- When requirements become clearer during exploration.

ADVANTAGE	Gives fine-grained control and supports continuous improvement.
------------------	---

LIMITATION	Many unfocused follow-ups can create contradictions or lose earlier requirements.
-------------------	---

COMMON MISTAKE	Saying only 'make it better' without defining what better means.
-----------------------	--

TRAINER NOTE	Use a quality checklist in each iteration: accuracy, clarity, audience, constraints and format.
---------------------	---

Quick Revision: Compare All 10 Techniques

Zero-shot	No example	Task + requirement	Direct tasks
One-shot	One example	Example + new input	Simple pattern copying
Few-shot	Several examples	Examples + new input	Classification/consistency
Role-based	Assign expertise	Role + task + audience	Expert tone/perspective
Context-based	Give background	Background + situation + task	Personalised answers
Instruction-based	Give actions	Steps + deliverable	Multi-step work
Constraint-based	Set limits	Task + rules + limits	Controlled output
Negative	State exclusions	Include + do not include	Avoid unwanted content
Output-format	Define structure	Task + fields/columns	Tables, JSON, reports
Iterative	Refine in turns	Draft + review + improve	Complex creation

EASY MEMORY

Examples • Exclusions • Act as a role • Mention context • Provide steps • Limits • Expected format • Step-by-step refinement

From Weak Prompt to Professional Prompt

A prompt becomes stronger when every added detail removes a meaningful uncertainty.

WEAK	Explain Java.
AUDIENCE	Explain Java to a beginner with no programming experience.
SCOPE	Explain variables, conditions, loops and methods in Java to a beginner.
METHOD	Explain each concept with one definition and one simple code example.
CONSTRAINTS	Use Java 17, simple English and fewer than 400 words. Avoid frameworks.
FORMAT	Present each concept under Definition, Syntax, Example and Practice Question.

Final combined prompt

PROMPT	Act as a patient Java trainer. Explain variables, conditions, loops and methods to a beginner with no programming experience. For each concept, provide a simple definition, syntax, one small Java 17 example and one practice question. Use simple English, avoid frameworks and keep the complete answer below 400 words. Present each concept under the headings Definition, Syntax, Example and Practice Question.
QUALITY CHECK	Is the task clear? Is the audience known? Is necessary context present? Are requirements measurable? Are exclusions useful? Is the output format explicit?

Student Practice Lab

Complete these activities individually. Test your prompts in an AI tool, review the answers and improve the prompts where necessary.

Task 1 • Zero-shot

Ask AI to write a professional leave email for two days.

Task 2 • One-shot

Give one example of informal-to-professional rewriting, then transform a new sentence.

Task 3 • Few-shot

Use three labelled examples to classify customer feedback as Positive, Negative or Neutral.

Task 4 • Role-based

Create a mock Java interview prompt for a developer with two years of experience.

Task 5 • Context-based

Provide class size, learner level and available time; ask for a Java training plan.

Task 6 • Instruction-based

Ask AI to review code using at least five numbered actions.

Task 7 • Constraint-based

Create a LinkedIn post prompt with measurable word, bullet and hashtag limits.

Task 8 • Negative

Explain AI to a child while excluding technical words, mathematics and code.

Task 9 • Output-format

Compare Java and Python using an exact four-column table.

Task 10 • Iterative

Create a resume summary and improve it through at least four follow-up prompts.

Capstone Assignment: Design a 5-Day Java Training Plan

Write ten separate prompts for the same goal - one prompt for each technique. Then create one final combined professional prompt.

FINAL RESULT MUST INCLUDE

Three topics per day • one practical program per day • one assignment per day • duration for every topic • beginner-friendly explanations • table format • maximum 500 words

Correct use of techniques	8	The prompt clearly demonstrates the named technique.
Clarity and completeness	4	Task, audience and important requirements are understandable.
Context and constraints	3	Relevant background and measurable limits are supplied.
Output quality	3	The expected response is useful, accurate and well structured.
Creativity	2	Examples are realistic, engaging and original.

SUBMISSION CHECKLIST

10 technique-specific prompts completed • examples tested • outputs reviewed • one weak prompt improved • final combined prompt included • reflection written in 100 words

One-Page Prompt Builder

Copy this template whenever you need to create a professional prompt.

ROLE: Act as a [specific role].

CONTEXT: [Who am I? Who is the audience? What is the situation?]

TASK: [State one clear goal using an action verb.]

INSTRUCTIONS: 1) [...] 2) [...] 3) [...]

MUST INCLUDE: [required facts, sections or examples]

CONSTRAINTS: [length, number, budget, time, technology, tone]

DO NOT INCLUDE: [irrelevant, sensitive or advanced content]

OUTPUT FORMAT: [table/JSON/bullets/email] with [exact fields or headings].

QUALITY CHECK: Verify [accuracy, completeness and audience suitability] before answering.

Final trainer notes

- Be specific, but include only relevant details.
- Use examples when a pattern is difficult to explain.
- Make limits measurable instead of vague.
- Tell the AI what to include as well as what to avoid.
- Treat the first answer as a draft for important work.
- Review AI output for facts, bias, privacy, safety and suitability.
- Never paste confidential organisational or personal data into an AI tool without authorisation.

THE CORE IDEA

Good prompting is not about using complicated English. It is about communicating the goal, situation, rules and expected result clearly.

Explore. Experiment. Evaluate. Improve.